

Academic Challenges Among University Students Through Mobile Technology

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Abstract—Mobile technology has greatly transformed the academic settings in higher learning institutions in Bangladesh because of its high adoption. Students at the university level have become very dependent on mobile phones and online services to attend their classes, get access to learning resources, submit their assignments, and most importantly to communicate with their teachers and lecturers. Although such technological integration has enhanced flexibility and accessibility, it has also posed serious academic challenges in the event of the absence of classes or technical issues like failure of the internet, malfunctioning of the devices or loss of data. This paper explores the character of these academic challenges and how the effects of the same can be mitigated using technological tools on mobiles. The study uses primary data gathered in a survey of 55 students via an online questionnaire to examine the trends in mobile technology usage, technical problem frequency and effects as well as how students rate digital solutions (online communication applications, cloud storage services, and centralized learning portals). The results indicate that mobile technology dependence is high, technical inconveniences are common, and that students strongly believe that the only way of continuing with studies is through technology based solutions. The paper is rounded off by making some recommendations on what can be done to improve digital infrastructure and support systems at educational institutions.

Keywords— mobile technology, academic challenges, university students, online learning platforms, cloud storage, educational portals

I. INTRODUCTION

A. Background information:

The mobile technology has turned out to be an important component of higher education across the globe. The accelerated growth of smart phones, cloud computing, and web based communication systems has made the traditional learning systems to be flexible and technology powered. Mobile learning means that students can have access to academic materials at any time or location, thus education is more inclusive and is more accommodating to needs of students [1]. Online lectures, assessment, and academic communication have been adopted more in universities in Bangladesh.

Mobile gadgets, including smartphones and laptops, are significantly used to support online courses, read lecture notes, assignments, and interact with peers and teachers by university students in Bangladesh. This dependence grew even bigger following the COVID-19 pandemic that moved most academic tasks online. Mobile technology has become more than an assistance device and an essential part of academic life consequently.

B. Overview

Mobile technology has presented a number of academic challenges despite its benefits. Late and absent students find it difficult to catch up with missed classes because of sickness, personal or technical reasons. Some of the most important lecture explanations, notes and announcements can be lost, leaving lapses in knowledge. Learning is often interrupted by technical problems like low internet connectivity, battery malfunctions, physical damage of devices or accidental data loss. It has been found that recurrent technological interruptions lower student involvement, raise stress levels, and adversely influence academic results [2].

These are even more damaging in developing economies such as Bangladesh whereby most students rely on a single machine to carry out all school related processes. Being denied temporary access to either a laptop or a phone may result in missed classes, submissions, and anxiety about school. Research that has been conducted on mobile learning indicates that unless digital support systems are put in place, overreliance on technology may rather make students more vulnerable in terms of their academic well being [3].

These challenges are solved at the same time by mobile technology solutions. Online communication platforms like Microsoft Teams, Messenger, and WhatsApp assist the students in gathering notes and information by other students. Google drive and One drive are cloud computing platforms where academic materials can be backed and recovered automatically. Instructional lectures, notices and even recorded classes can be accessed on any device, with security measures, in centralized learning portals. It has been found that such solutions are essential to the continuity of academics in cases of technical disruption [4].

This paper will discuss the problem of mobile among university students in Bangladesh and how the mobile based technological solutions can alleviate these problems and enhance academic performance of students.

II. METHODOLOGY

A. Research Design:

This study used a quantitative research approach and collected primary data from students through an online structured questionnaire. The primary aim of the methodology was to obtain the measurable data on how students use mobile technology, what academic problems they have because of the missed classes and technical failures, and what they consider to be the potential solutions to the problem technology related.

B. Sample Selection:

A total of 55 respondents were used to gather data. The respondents were students of HSC, undergraduate and postgraduate. The majority of the respondents were undergraduate students because they are the most active users of mobile technology in respect to academic processes. The participants have been selected due to the frequency of attending online classes, submitting assignments online, and the use of communication platforms to interact academically. Students in the other groups (school level) were excluded since their academic activities do not require as much of independent mobile based learning systems. The survey questionnaire had about 12-15 close ended questions. It was constructed to gather data in a few major areas, such as demographic information, the use of the devices, how frequently they used the mobile technology, what challenges they face at school due to technological problems, and views on the digital solutions. The questionnaire was shared through Google Forms that is easy to access and automatically gathers data.

C. Data Collection:

The questionnaire was distributed on social networks, including Facebook, WhatsApp, Instagram, and Telegram, in order to reach a large audience of students. The reason behind using this method of distribution is that university students in Bangladesh can be found to be very active in these sites. The data collection took place in one week and this allowed timely responses and at the same time the data was relevant. The analysis of the responses was done by analyzing the data using percentage analysis after data collection. The data were represented in charts and graphs that were created with the help of Google Forms. These visual aids were used to determine trends and patterns regarding the use of mobile technology, academic issues, and the usefulness of technology based interventions.

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III. RESULTS AND DISCUSSION

The survey findings offer an in depth account of the ways of studying with the help of mobile technology and the problems that the students of the University of Bangladesh encounter because of technical problems and missed lectures.

The demographic analysis has indicated that the majority of the respondents are undergraduate students, which represents the most active group of mobile based learning. The gender distribution shows that there are more male students involved though the female students are also a considerable

percentage of the sample. These demographic trends give valuable background of the findings.

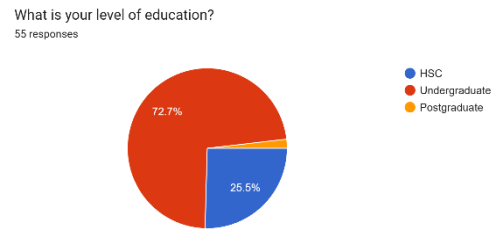


Fig. 1. Level of Education of Respondents

The device usage analysis shows that smartphones are mostly used in academics after laptops. This observation underscores the portability and affordability of smartphones that have become the most important academic tool of most students. A lower percentage of respondents use desktop computers probably because of their little mobility.

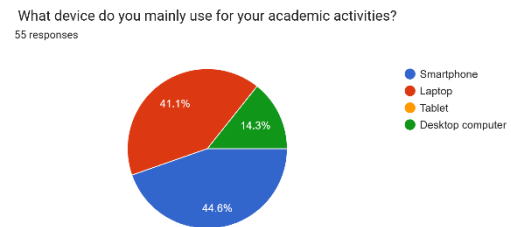


Fig. 2. Primary Device Used for Academic Activities

The rate of usage of mobile technologies also proves that students rely on digital tools. A big percentage of the respondents have been found to use mobile technology very often or quite often with respect to academics. It means that mobile devices have become inseparable parts of day to day academic processes, the process of attending classes, taking notes, and fulfilling assignments. This degree of dependence is consistent with available literature on the increased importance of mobile learning in higher education [1].

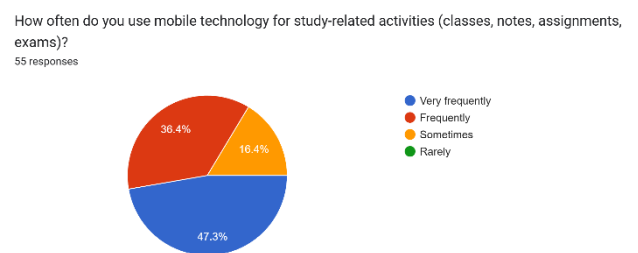


Fig. 3. Frequency of Mobile Technology Use

As regards academic communication, the students indicated that they used various online platforms. The most popular platform that was used was Microsoft Teams due to its adoption by universities in formal academic activities. There was also widespread use of WhatsApp and Messenger, mostly used in informal communication and peer support. Such applications enable learners to exchange notes, announcements, and updates fast especially when classes are missed. Earlier research has indicated that the use of online communication tools has a very great impact on the

development of teamwork and exchange of information among students [3].

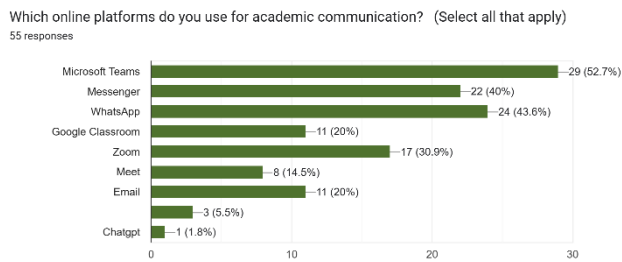


Fig. 4. Online Platforms Used for Academic Communication

One of the frequent problems was reported to be technical issues. Interruptions related to the failure of the internet, as well as device malfunction and battery issues, were experienced by at least half of the students. Very little percentage indicated that they never had such problems. Technical breaks are frequent and this may disrupt learning and may cause frustration and stress that adversely affect academic engagements and performance [2].

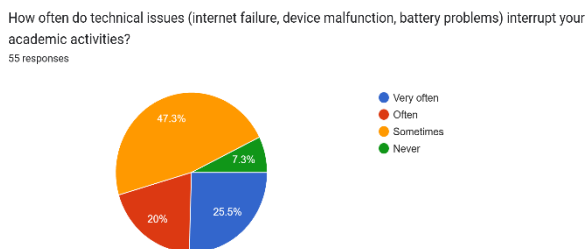


Fig. 5. Frequency of Technical Issues Interrupting Academic Activities

Students find themselves in a number of academic trouble in case they skip classes. The problem that was most reported was the inability to understand what was taught later on, with the second issue being the lack of lecture notes. Being late on critical announcements and feeling stressed during exams were also not uncommon. These results indicate that not attending classes has not just the immediate learning but could also lead to long term knowledge gaps that influence exam scores and academic self confidence.

The problem of data loss became a major concern whereby a substantial number of students reported to have lost academic materials either through phone damage, loss, or technical failure. This brings out the dangers of keeping academic information in one machine. It is highlighted in the research that cloud based backup systems are necessary to secure academic documentation as well as guarantee data recovery [4].

These issues were evident in the responses in terms of their effects on academic performance. A majority of the students have claimed that the technical problems and missed materials intermediately or critically influenced their academic outcomes. This proves the fact that there are actual and quantifiable impacts of mobile related issues on learning performance and academic achievement [2].

How much do these problems affect your academic performance?
55 responses

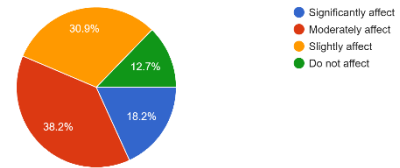


Fig. 6. Impact on Academic Performance

The use of technology based solutions among the students was also studied. The majority of the respondents said that they frequently gathered classroom notes and information via online communication applications through the exchange with classmates. These tools were considered to be very effective or effective in assisting the students to reclaim lost academic materials. This observation is in line with the notion that peer based online communication is an important aspect of academic recovery [3].

How effective are online communication apps (Teams, Messenger, WhatsApp) in helping you recover missed academic information?
55 responses

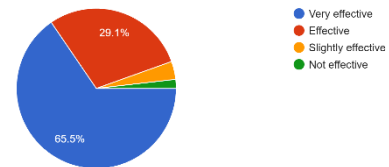


Fig. 7. Effectiveness of Communication Apps

There was also high usage of cloud storage by the students. The service that was most utilized was Google drive, which was then OneDrive and Dropbox. The option of using cloud storage on a regular basis would enable students to automatically save academic content and view it anywhere, thereby minimizing the chances of losing access to it forever.

How regularly do you use cloud storage services to back up your academic materials?
55 responses

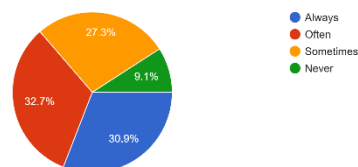


Fig. 8. Frequency of Cloud Storage Usage

Lastly, nearly all the respondents maintained that a centralized educational portal would also be useful in achieving continuity of academic processes in the event of technical issues. These portals have the ability to incorporate lectures, announcements, recordings and submissions in a single system that is safe. There is solid research that confirms the ease of centralized digital platforms in enhancing the continuity of learning and minimizing academic interruption [3].

Which mobile based technology feature helps you the most in reducing academic challenges caused by technical issues?
55 responses

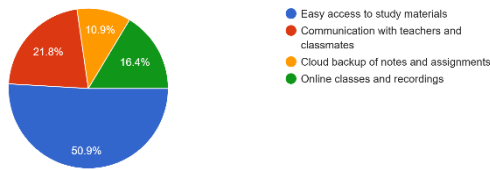


Fig. 9. Centralizing Educational Portal Helpfulness

However, in spite of its advantages, there are limitations of the study. The sample is not too large, and the statistics is self reported, which is also subject to personal bias. A bigger sample and qualitative interviews can be a part of the future studies in order to get deeper insights.

The results of this research have significant implications on the education schools and policy makers in Bangladesh. The fact that mobile technology has a high dependency rate among the students of the university shows that digital learning has ceased to be an added benefit, but a necessity. Hence, colleges should acknowledge mobile based learning as a fundamental part of the academic facilities and not a supportive tool. Without institutional planning, the students are prone to academic interferences due to technical breakdown, absence of classes, or loss of data.

On the whole, the results focus on the fact that students cannot solve academic problems concerning mobile technology. The planning of institutions, providing digital policies, and putting up infrastructure are needed to make the academic system robust and inclusive.

IV. LIMITATIONS OF THE STUDY

This study has numerous weaknesses even though it does offer useful insights. To begin with, the sample was only 55 respondents, which might not be very representative of the experiences of all the university students in Bangladesh. Second, the data was gathered as self reported answers to surveys, and this can be subjected to personal bias or incorrect data. Third, the researchers majorly considered the opinions of the students and did not involve teachers or university administrators. The research could be improved by future studies to consider the limitations by applying a bigger sample size, interviews based on qualitative analysis, and other perspectives of stakeholders.

V. RECOMMENDATION

According to the results of this research, it is possible to make a number of recommendations. Learning institutions must enhance their online infrastructure so that they can have seamless access to learning materials. Any universities are advised to embrace cloud based backup systems amongst students to minimize the chances of information loss. Moreover, it is highly advisable to develop integrated academic portals which can be accessed by students to access lecture materials, announcements, and recorded classes and assessments. Students can also be supported by the training programs on effective usage of mobile technology and digital resource management. With all these in place, the universities are able to reduce the academic disturbance and generate a more resilient and accommodating learning environment.

VI. CONCLUSION

In this paper, the issues of academic challenges encountered by students at the university, in Bangladesh, because of their growing reliance on mobile technology. The results show that even though mobile based learning is flexible and accessible, it also poses a serious risk to students in case of technical problems or the lack of classes. Technical interruptions, loss of data, and inaccessibility to academic material are among the problems that have adverse impacts on the academic performance of students and cause them to be stressed with academic issues. These outcomes affirm that lack of proper system to support technological dependence may interfere with continuity of learning.

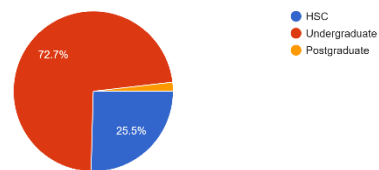
Concurrently, the paper illustrates that mobile technology by itself can provide useful remedies to these predicaments. Centralized education portals, online communication systems, and cloud storage systems are critical towards ensuring continuity of academics. The results indicate that with the correct use of these tools, students will be able to recount the lost academic information as well as mitigate the adverse effects of the technical inconveniences.

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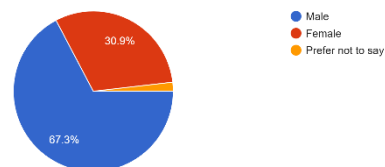
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- [5] D. G. Oblinger and J. L. Oblinger, "Educating the Net Generation," EDUCAUSE, 2005.

APPENDIX

What is your level of education?
55 responses

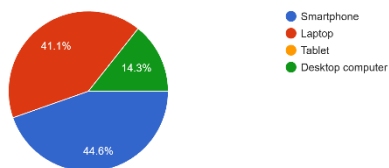


What is your gender?
55 responses



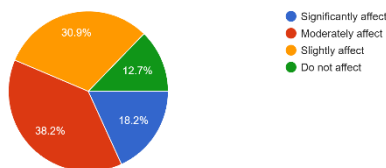
What device do you mainly use for your academic activities?

55 responses



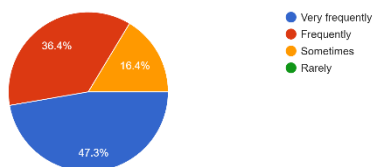
How much do these problems affect your academic performance?

55 responses



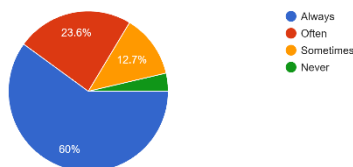
How often do you use mobile technology for study-related activities (classes, notes, assignments, exams)?

55 responses



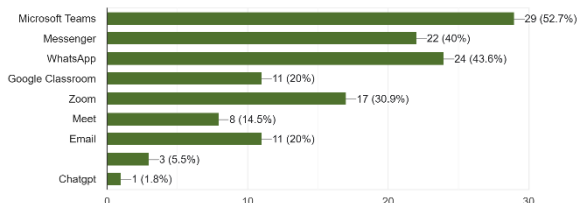
Do you collect class notes or information from classmates using online communication apps?

55 responses



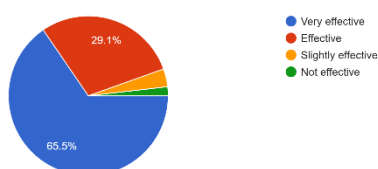
Which online platforms do you use for academic communication? (Select all that apply)

55 responses



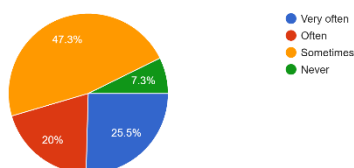
How effective are online communication apps (Teams, Messenger, WhatsApp) in helping you recover missed academic information?

55 responses



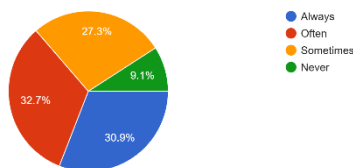
How often do technical issues (internet failure, device malfunction, battery problems) interrupt your academic activities?

55 responses



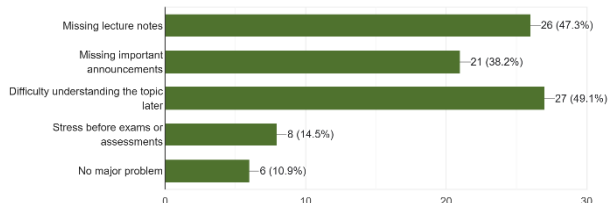
How regularly do you use cloud storage services to back up your academic materials?

55 responses



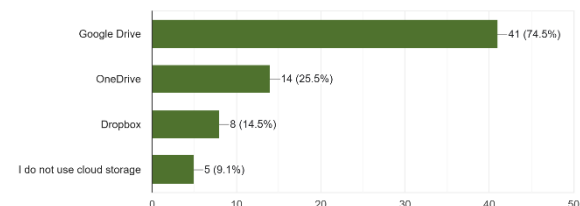
When you miss a class, what problems do you usually face? (Select all that apply)

55 responses



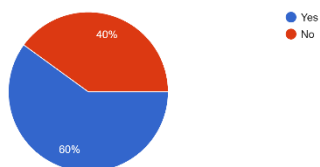
Which cloud storage services do you use for saving academic materials? (Select all that apply)

55 responses



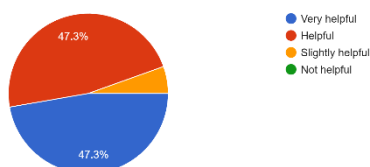
Have you ever lost academic materials (notes, assignments, recordings) because of phone damage, loss, or technical failure?

55 responses

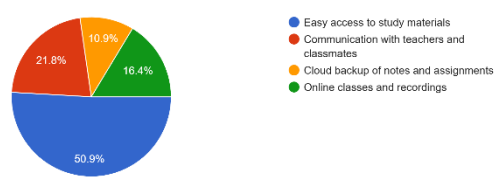


How helpful would a centralized educational portal be in ensuring academic continuity during technical problems?

55 responses



Which mobile based technology feature helps you the most in reducing academic challenges caused by technical issues?
55 responses



In your opinion, what additional support should educational institutions provide to help students facing technical problems?
55 responses

